📡 Configuración del Sistema RFID + MQTT

Parametrización de los lectores

🔗 Conexión MQTT

- **Broker:** tcp://192.168.1.155:1883
- **Usuario:** gestion_rfid
- **Password:** rfid_room@

🏢 Configuración Lectores – ALMACÉN

Lector 1 – IP: 192.168.1.158

Payload (JSON / MQTT):

```
[{
  "event": "TAG_READ",
  "topic": "TAG_readings",
  "body": "var body = '{';
          body += '\"devid\": \"Almacen-1\",';
          body += '\"devip\": \"' + ctx_devip + '\",';
          body += '\"devmac\": \"' + ctx_devmac + '\",';
          body += '\"reads\": [';
          for(i = 0; i < ctx_tagno; i++) {
            body += '{';
            body += '\"EPC\": \"' + ctx_tags[i].getEPC() + '\",';
            body += '\"Antenna\": \"' + ctx_tags[i].getAntenna() + '\",';
            body += '\"RSSI\": \"' + ctx_tags[i].getRSSI() + '\"';
            body += '}';
            if(i < ctx_tagno - 1) { body += ','; }
          }
          body += ']';
          body += '}';"
}]
```

Lector 2 – IP: 192.168.1.157

Payload (JSON / MQTT):
(idéntico al anterior, ajustando solo los metadatos de contexto).

⚙️ Opciones de envío MQTT (Lectores Almacén)

- **Send one by one:** ✕ No conectado
- **Inventory tag TTL (s):** 15
- **Keep-alive period (ms):** -1 (deshabilitado)

- **Use antenna in tagTTL:** ☑ Conectado
- **Re-send on COMM error:** ✕ No conectado

🖥 Configuración Lector – PUERTA (con alarma, EPC filtrado y eventos)

Lector – IP: 192.168.1.159

Payload (JSON / MQTT):

```
[{
  "event": "TAG_READ",
  "topic": "TAG_readings",
  "body": "var body = '{';
          body += '\"devid\": \"Puerta-1\",';
          body += '\"devip\": \"' + ctx_devip + '\",';
          body += '\"devmac\": \"' + ctx_devmac + '\",';
          body += '\"reads\": [';
          for(i = 0; i < ctx_tagno; i++) {
            body += '{';
            body += '\"EPC\": \"' + ctx_tags[i].getEPC() + '\",';
            body += '\"Antenna\": \"' + ctx_tags[i].getAntenna() + '\",';
            body += '\"RSSI\": \"' + ctx_tags[i].getRSSI() + '\"';
            body += '}';
            if(i < ctx_tagno - 1) { body += ','; }
          }
          body += ']';
          body += '}';"
}]
```

⚙ Opciones de envío MQTT (Puerta)

- **Send one by one:** ✕ No conectado
- **Inventory tag TTL (s):** 15
- **Keep-alive period (ms):** -1 (deshabilitado)
- **Use antenna in tagTTL:** ☑ Conectado
- **Re-send on COMM error:** ✕ No conectado

🎯 Configuración avanzada – PUERTA (Filtrado EPC + Base de Datos)

🔍 Filtro EPCGen2

- **Mask offset (bits):** 32
- **Filter mask (hex):** 0xce002812
- **Mask length (bits):** 16
- **Use SW filter only:** ✕ Desconectado *(puede degradar rendimiento en lecturas intensivas)*

- **Allow Extended EPC Filter:** ✕ Desconectado

🖴 Configuración de Alarma + DB

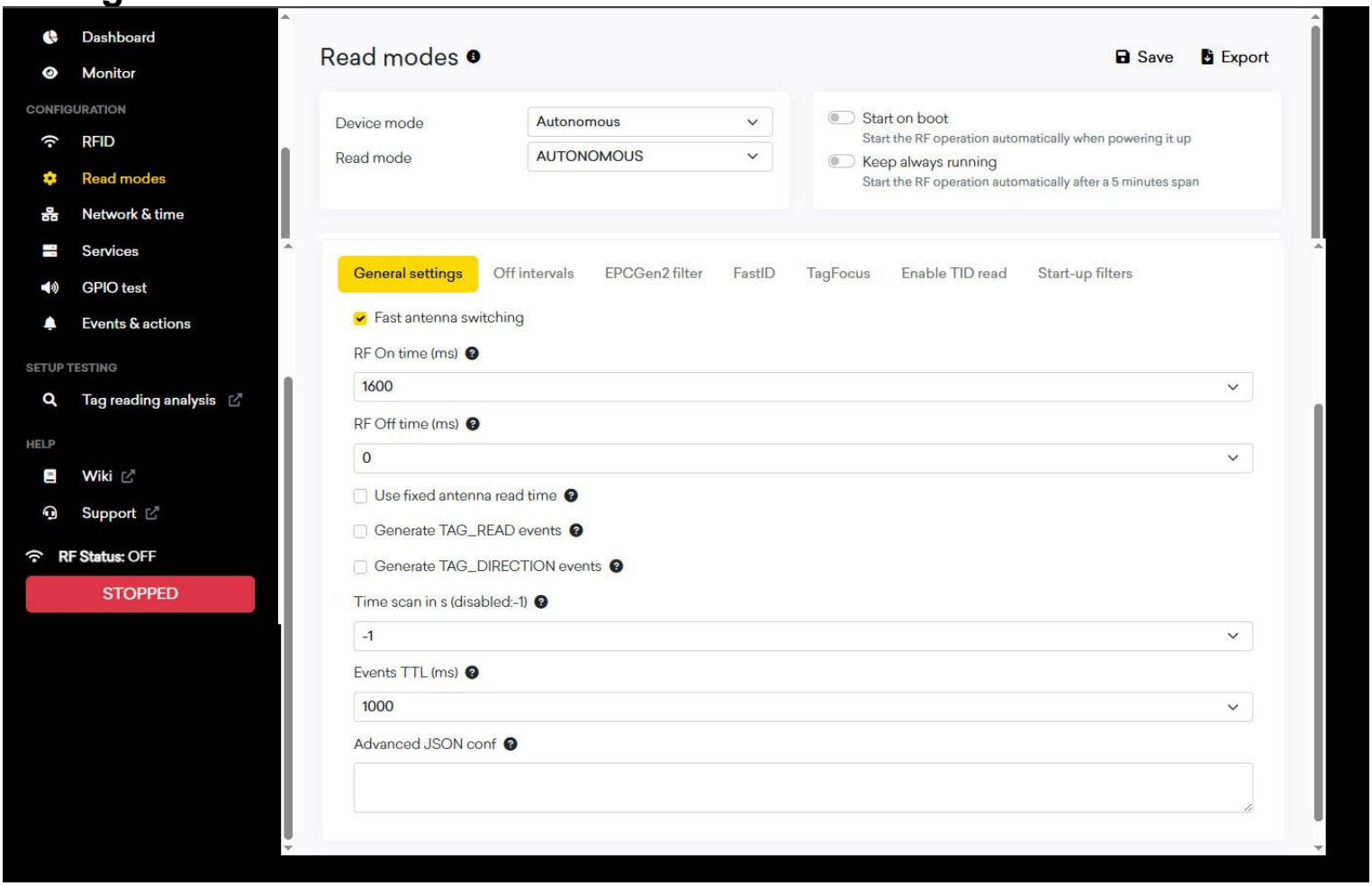
- **Java Driver Class:** org.postgresql.Driver
- **Connection String:** jdbc:postgresql://192.168.1.155:5432/gestion_rfid
- **DB User:** rfidroom
- **DB Password:** rfidroom@2025

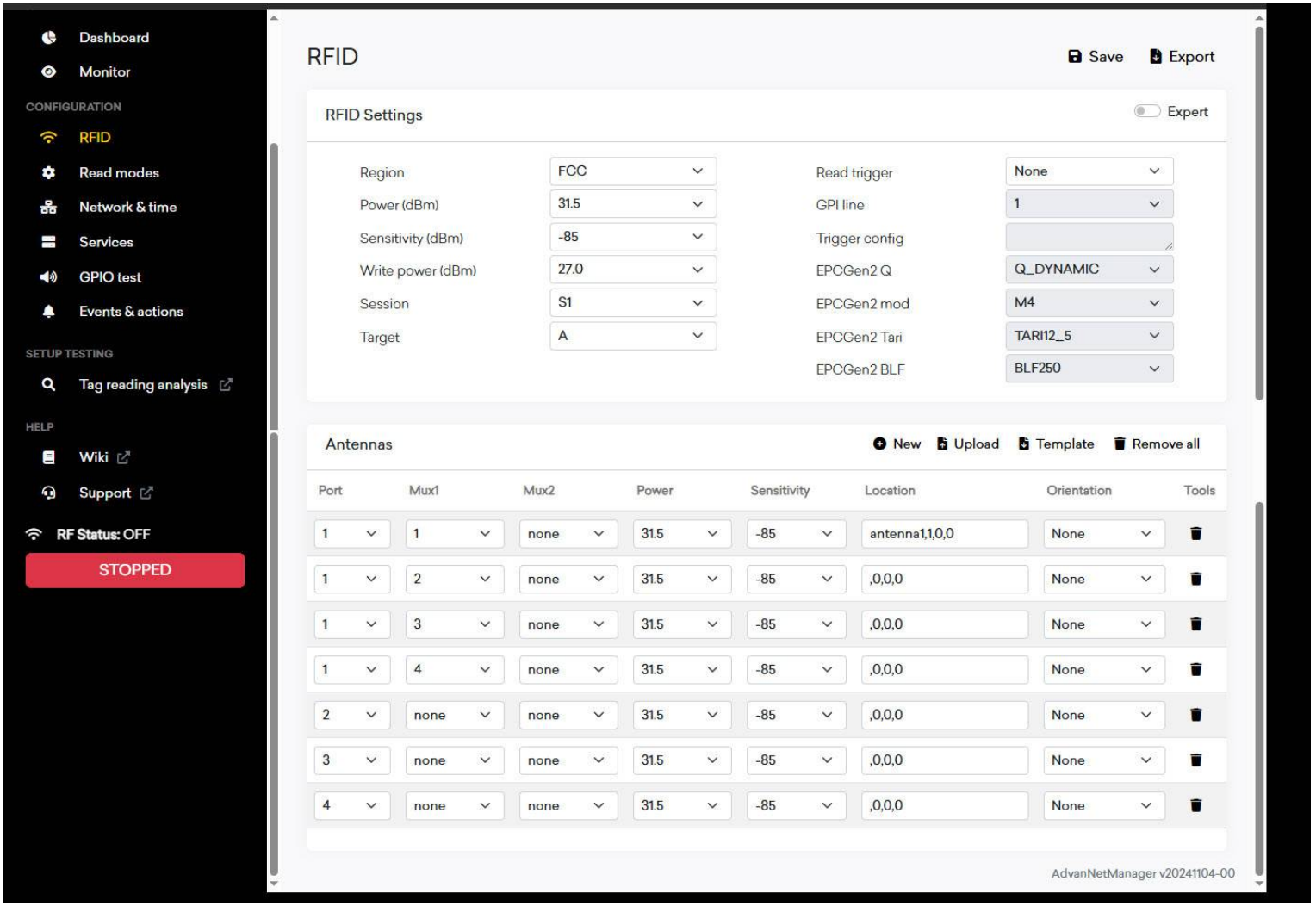
Query de validación:

```
SELECT "state"
FROM "user_room"
WHERE "epc" = ${epc}
  AND "state" = 1
  AND "door" = 'Puerta-1';
```

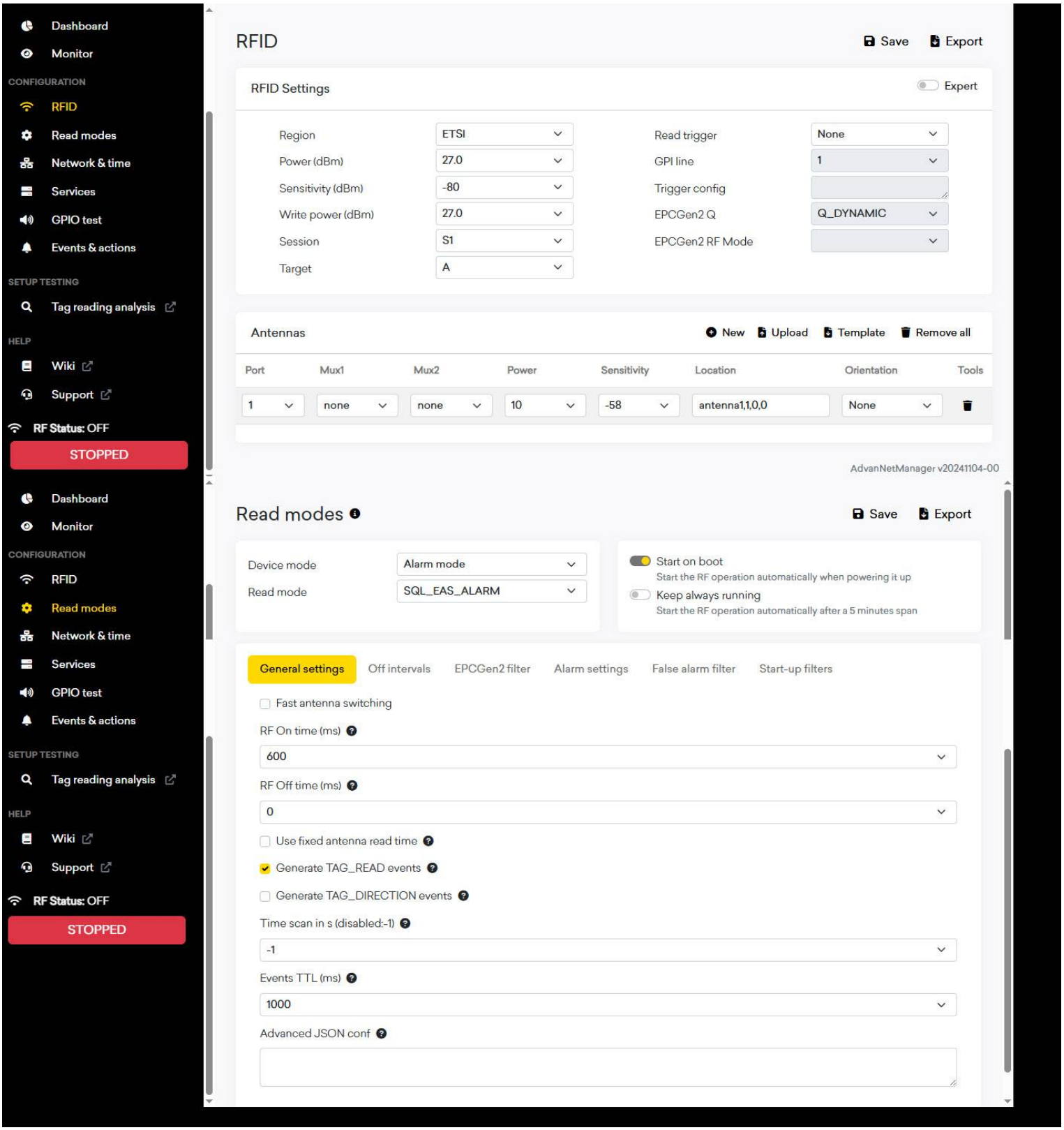
- **Invert query result:** ✕ Desconectado
- **Query cache TTL (ms):** 1500

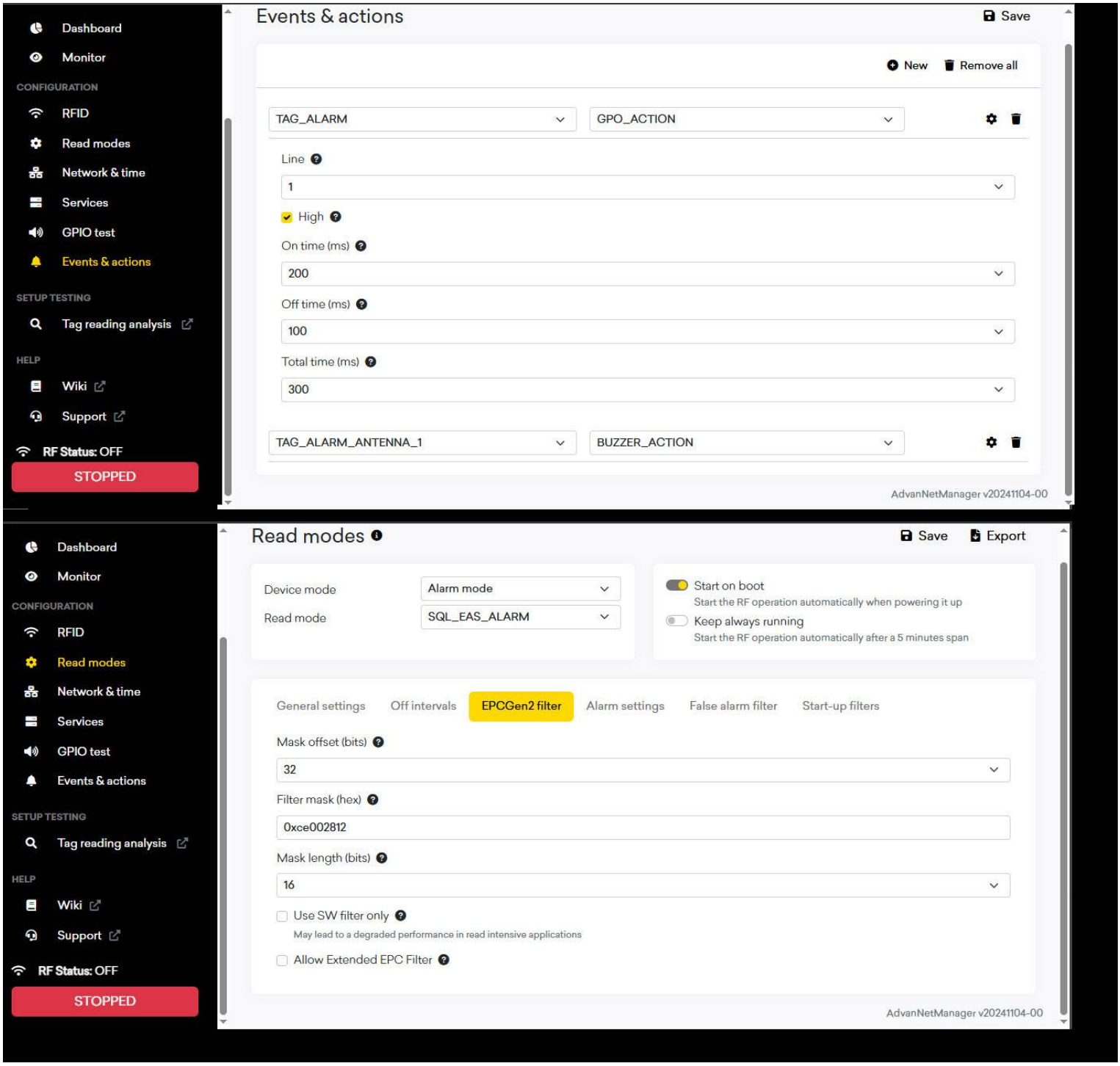
Configuratión WEB Lectores ALMACEN.




Dashboard
Monitor
CONFIGURATION
RFID
Read modes
Network & time
Services
GPIO test
Events & actions
SETUP TESTING
Tag reading analysis
HELP
Wiki
Support
RF Status: OFF
STOPPED
RFID
Save
Export
RFID Settings
Expert
Region
FCC
Power (dBm)
31.5
Sensitivity (dBm)
-85
Write power (dBm)
27.0
Session
S1
Target
A
Read trigger
None
GPI line
1
Trigger config
EPCGen2 Q
Q_DYNAMIC
EPCGen2 mod
M4
EPCGen2 Tari
TARI12_5
EPCGen2 BLF
BLF250
Antennas
New
Upload
Template
Remove all
Port
Mux1
Mux2
Power
Sensitivity
Location
Orientation
Tools
1 1 none 31.5 -85 antenna1,1,0,0 None
1 2 none 31.5 -85 ,0,0,0 None
1 3 none 31.5 -85 ,0,0,0 None
1 4 none 31.5 -85 ,0,0,0 None
2 none none 31.5 -85 ,0,0,0 None
3 none none 31.5 -85 ,0,0,0 None
4 none none 31.5 -85 ,0,0,0 None
AdvanNetManager v20241104-00

Configuratión WEB Lector PUERTA.




Dashboard
Monitor
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GPIO test
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Tag reading analysis
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Support
RF Status: OFF
STOPPED
Events & actions
Save
New
Remove all
TAG_ALARM
GPO_ACTION
Line
1
High
On time (ms)
200
Off time (ms)
100
Total time (ms)
300
TAG_ALARM_ANTENNA_1
BUZZER_ACTION
AdvanNetManager v20241104-00

Dashboard
Monitor
CONFIGURATION
RFID
Read modes
Network & time
Services
GPIO test
Events & actions
SETUP TESTING
Tag reading analysis
HELP
Wiki
Support
RF Status: OFF
STOPPED
Read modes
Save
Export
Device mode
Alarm mode
Read mode
SQL_EAS_ALARM
Start on boot
Start the RF operation automatically when powering it up
Keep always running
Start the RF operation automatically after a 5 minutes span
General settings
Off intervals
EPCGen2 filter
Alarm settings
False alarm filter
Start-up filters
Mask offset (bits)
32
Filter mask (hex)
0xce002812
Mask length (bits)
16
Use SW filter only
May lead to a degraded performance in read intensive applications
Allow Extended EPC Filter
AdvanNetManager v20241104-00